

Module 1: Introduction and Essentials of Python

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Learning Objectives

- Introduction to Python programming
- Basic Python syntax and data types
- Variables, operators, and expressions
- Control flow (if/else, loops)
- Functions and modules

Topics Covered

1. Python installation and setup
2. Python interpreter and IDEs
3. Basic data types (int, float, string, boolean)
4. Variables and assignment
5. Arithmetic and comparison operators
6. Conditional statements
7. Loops (for and while)
8. Functions definition and calling
9. Importing modules
10. Basic error handling

Exercises

- Practice problems will be provided throughout the module

1. Python Installation and Setup

In [1...

```
1 # Check Python version
2 import sys
3 print(f"Python version: {sys.version}")
4 print(f"Python executable: {sys.executable}")
5
```

Python version: 3.11.13 (main, Jun 5 2025, 08:14:07) [Clang 14.0.6]
Python executable: /opt/miniconda3/envs/python-course/bin/python

2. Basic Data Types

In [3...

```
1      # Integer
2      age = 25
3      print(f"Age: {age}, Type: {type(age)}")
4
5      # Float
6      height = 5.9
7      print(f"Height: {height}, Type: {type(height)}")
8
9      # String
10     name = "Alice"
11     print(f"Name: {name}, Type: {type(name)}")
12
13     # Boolean
14     is_student = True
15     print(f"Is student: {is_student}, Type: {type(is_student)}")
16
17     # None
18     data = None
19     print(f"Data: {data}, Type: {type(data)}")
20
```

Age: 25, Type: <class 'int'>
Height: 5.9, Type: <class 'float'>
Name: Alice, Type: <class 'str'>
Is student: True, Type: <class 'bool'>
Data: None, Type: <class 'NoneType'>

3. Variables and Assignment

In [4...

```
1      # Variable assignment
2      x = 10
3      y = 20
4
5      # Multiple assignment
6      a, b, c = 1, 2, 3
7      print(f"a={a}, b={b}, c={c}")
8
9      # Swapping variables
10     x, y = y, x
11     print(f"After swap: x={x}, y={y}")
12
13     # Augmented assignment
14     count = 5
15     count += 3 # Same as count = count + 3
16     print(f"Count after += 3: {count}")
17
18     # String formatting
19     first_name = "John"
20     last_name = "Doe"
21
22
23
```

```
full_name = f"{first_name} {last_name}"
print(f"Full name: {full_name}")
```

a=1, b=2, c=3
After swap: x=20, y=10
Count after += 3: 8
Full name: John Doe

4. Arithmetic and Comparison Operators

In [5...

```
1  # Arithmetic operators
2
3  a, b = 10, 3
4
5  print(f"Addition: {a} + {b} = {a + b}")
6  print(f"Subtraction: {a} - {b} = {a - b}")
7  print(f"Multiplication: {a} * {b} = {a * b}")
8  print(f"Division: {a} / {b} = {a / b}")
9  print(f"Floor division: {a} // {b} = {a // b}")
10 print(f"Modulus: {a} % {b} = {a % b}")
11 print(f"Exponentiation: {a} ** {b} = {a ** b}")
12
13 # Comparison operators
14 x, y = 5, 8
15 print(f"\nComparison operators:")
16 print(f"{x} == {y}: {x == y}")
17 print(f"{x} != {y}: {x != y}")
18 print(f"{x} < {y}: {x < y}")
19 print(f"{x} > {y}: {x > y}")
20 print(f"{x} <= {y}: {x <= y}")
21 print(f"{x} >= {y}: {x >= y}")
22
23 # Logical operators
24 p, q = True, False
25 print(f"\nLogical operators:")
26 print(f"p and q: {p and q}")
27 print(f"p or q: {p or q}")
28 print(f"not p: {not p}")
```

Addition: $10 + 3 = 13$
Subtraction: $10 - 3 = 7$
Multiplication: $10 * 3 = 30$
Division: $10 / 3 = 3.3333333333333335$
Floor division: $10 // 3 = 3$
Modulus: $10 \% 3 = 1$
Exponentiation: $10 ** 3 = 1000$

Comparison operators:

5 == 8: False
5 != 8: True
5 < 8: True
5 > 8: False
5 <= 8: True
5 >= 8: False

Logical operators:

p and q: False
p or q: True
not p: False

5. Conditional Statements

In [6...

```
1
2  # if-else statements
3  score = 85
4
5  if score >= 90:
6      grade = "A"
7  elif score >= 80:
8      grade = "B"
9  elif score >= 70:
10     grade = "C"
11  elif score >= 60:
12     grade = "D"
13  else:
14     grade = "F"
15
16  print(f"Score: {score}, Grade: {grade}")
17
18  # Ternary operator
19  age = 20
20  status = "adult" if age >= 18 else "minor"
21  print(f"Age: {age}, Status: {status}")
22
23  # Nested conditions
24  temperature = 25
25  is_sunny = True
26
27
28  if temperature > 20:
29      if is_sunny:
30          print("Great weather for outdoor activities!")
31      else:
32          print("Warm but cloudy day")
33  else:
34      print("It's a bit cold today")
```

Score: 85, Grade: B
Age: 20, Status: adult
Great weather for outdoor activities!

6. Loops

In [7...

```
1  # for loop
2  fruits = ["apple", "banana", "orange", "grape"]
3  print("Fruits:")
4  for fruit in fruits:
5      print(f"- {fruit}")
6
7
8  # for loop with range
9  print("\nNumbers 1 to 5:")
10 for i in range(1, 6):
11     print(f"Number: {i}")
12
13 # while loop
14 count = 0
15 print("\nWhile loop countdown:")
16 while count < 5:
17     print(f"Count: {count}")
18     count += 1
19
20
21 # break and continue
22 print("\nBreak and continue example:")
23 for num in range(1, 11):
24     if num == 3:
25         continue # Skip 3
26     if num == 8:
27         break # Stop at 8
28     print(f"Number: {num}")
29
30
31 # enumerate
32 print("\nEnumerate example:")
33 for index, fruit in enumerate(fruits):
34     print(f"{index}: {fruit}")
```

Fruits:
– apple
– banana
– orange
– grape

Numbers 1 to 5:
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5

While loop countdown:
Count: 0
Count: 1
Count: 2
Count: 3
Count: 4

Break and continue example:
Number: 1
Number: 2
Number: 4
Number: 5
Number: 6
Number: 7

Enumerate example:
0: apple
1: banana
2: orange
3: grape

7. Functions

In [8...

```
1
2  # Basic function
3  def greet(name):
4      return f"Hello, {name}!"
5
6  print(greet("Alice"))
7
8  # Function with multiple parameters
9  def calculate_area(length, width):
10     return length * width
11
12  area = calculate_area(5, 3)
13  print(f"Area: {area}")
14
15  # Function with default parameters
16  def greet_with_title(name, title="Mr."):
17      return f"Hello, {title} {name}!"
18
19  print(greet_with_title("Smith"))
20  print(greet_with_title("Johnson", "Dr.))
21
22
23  # Function with keyword arguments
```

```

24     def create_profile(name, age, city="Unknown", country="Unknown"):
25         return f"Name: {name}, Age: {age}, City: {city}, Country: {co
26
27     profile = create_profile("Bob", 30, country="USA")
28     print(profile)
29
30     # Function returning multiple values
31     def get_name_parts(full_name):
32         parts = full_name.split()
33         return parts[0], parts[-1] if len(parts) > 1 else ""
34
35     first, last = get_name_parts("John Doe")
36     print(f"First: {first}, Last: {last}")
37
38     # Lambda function
39     square = lambda x: x ** 2
40     print(f"Square of 5: {square(5)}")
41
42     # Using lambda with built-in functions
43     numbers = [1, 2, 3, 4, 5]
44     squared_numbers = list(map(lambda x: x ** 2, numbers))
45     print(f"Squared numbers: {squared_numbers}")

```

```

Hello, Alice!
Area: 15
Hello, Mr. Smith!
Hello, Dr. Johnson!
Name: Bob, Age: 30, City: Unknown, Country: USA
First: John, Last: Doe
Square of 5: 25
Squared numbers: [1, 4, 9, 16, 25]

```

8. Importing Modules

In [9...

```

1     # Import entire module
2     import math
3     print(f"Square root of 16: {math.sqrt(16)}")
4     print(f"Value of pi: {math.pi}")
5
6     # Import specific functions
7     from datetime import datetime, date
8     now = datetime.now()
9     today = date.today()
10    print(f"Current time: {now}")
11    print(f"Today's date: {today}")
12
13    # Import with alias
14    import random as rnd
15    random_number = rnd.randint(1, 10)
16    print(f"Random number: {random_number}")
17
18    # Import all from module (not recommended for large modules)
19    from math import *
20    print(f"Factorial of 5: {factorial(5)}")
21
22
23
24

```

```
# Check available functions in a module
print(f"\nMath module functions: {dir(math)[:10]}...") # First 1
```

Square root of 16: 4.0
Value of pi: 3.141592653589793
Current time: 2025-09-07 21:28:24.335830
Today's date: 2025-09-07
Random number: 3
Factorial of 5: 120

Math module functions: ['__doc__', '__file__', '__loader__', '__name__', '__package__', '__spec__', 'acos', 'acosh', 'asin', 'asinh']...

9. Basic Error Handling

In [1...

```
1  # Basic try-except
2  def safe_divide(a, b):
3      try:
4          result = a / b
5          return result
6      except ZeroDivisionError:
7          return "Cannot divide by zero!"
8      except TypeError:
9          return "Invalid input types!"
10
11  print(safe_divide(10, 2))
12  print(safe_divide(10, 0))
13  print(safe_divide(10, "2"))
14
15  # Multiple exception types
16  def get_list_element(lst, index):
17      try:
18          return lst[index]
19      except IndexError:
20          return "Index out of range!"
21      except TypeError:
22          return "Invalid index type!"
23
24  numbers = [1, 2, 3, 4, 5]
25  print(get_list_element(numbers, 2))
26  print(get_list_element(numbers, 10))
27  print(get_list_element(numbers, "2"))
28
29  # try-except-else-finally
30  def process_file(filename):
31      try:
32          file = open(filename, 'r')
33          content = file.read()
34      except FileNotFoundError:
35          print(f"File {filename} not found!")
36          return None
37      except PermissionError:
38          print(f"Permission denied to read {filename}")
39          return None
40      else:
41          print("File read successfully!")
```



```
45         return content
46     finally:
47         print("File processing completed")
48
49     # This will show the error handling
50     result = process_file("nonexistent.txt")
```

5.0

Cannot divide by zero!

Invalid input types!

3

Index out of range!

Invalid index type!

File nonexistent.txt not found!

File processing completed

In [...

```
1
2
```